Sharp parameter ranges for zero-balanced hypergeometric ratio inequalities MathDB 364530

Principia Math in Progress

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Task 1: Solve the original problem. The mathematical displays in the MathDB statement are incomplete. We use the precise source, Qiu–Ma–Xiang [1, Conjecture 6.2, p. 22], together with its equations (1.6)–(1.10) on page 2. For $a, b > 0$ and $0 < r < 1$, define

$$f_b(r) = \frac{{}_2F_1(a, b; a+b+1; r)}{{}_2F_1(a, b; a+b; r)}.$$

The parameter a is fixed unless stated otherwise. Euler's transformation gives

$$\frac{{}_2F_1(a+1, b+1; a+b+1; r)}{{}_2F_1(a, b; a+b; r)} = \frac{f_b(r)}{1-r}.$$

Thus the source asks for the sharp ranges for $\partial_b f_b(r) < 0$ and $f_{1-a}(r) > f_{3/2-a}(r)$ throughout $0 < r < 1$. The unrestricted earlier Baricz conjecture is already disproved in [1]; our subject is its corrected parameter-range conjecture.

Theorem 1. *For $a, b > 0$,*

$$\partial_b f_b(r) < 0 \text{ for every } r \in (0,1) \quad \Longleftrightarrow \quad b^2 \le a(a+1). \tag{1}$$

More generally, for $a > 0$ and $0 < b_1 < b_2$,

$$f_{b_1}(r) > f_{b_2}(r) \text{ for every } r \in (0,1) \quad \Longleftrightarrow \quad b_1 b_2 \le a(a+1). \tag{2}$$

Consequently, for $0 < a < 1$,

$$f_{1-a}(r) > f_{3/2-a}(r) \text{ for every } r \in (0,1) \quad \Longleftrightarrow \quad a \in [3/7, 1).$$

Both parts of source Conjecture 6.2 hold, including their boundary values.

The differential equation. Put $c = a + b$, $A = ab/c$, and $H(r) = {}_2F_1(a, b; c; r)$. The positive series and the larger denominator parameter in the numerator show that $0 < f_b(r) < 1$, with $f_b(0) = 1$. The derivative formula and Euler's transformation imply

$$\frac{H'(r)}{H(r)} = \frac{A}{1-r} f_b(r).$$

Substitution into the hypergeometric differential equation

$$r(1-r)H'' + [c - (c+1)r]H' - abH = 0$$

gives the Riccati equation

$$rf_b' = c(1 - f_b) - \frac{Ar}{1-r} f_b^2. \tag{3}$$

All functions involved extend analytically to $r = 0$. Their parameter derivatives are legitimate on compact subsets of the positive parameter domain and $0 \le r < 1$, by local uniform convergence of the hypergeometric series.

A positivity principle at the singular endpoint. We will repeatedly use the following elementary observation. If $c > 0$, q is continuous at zero and on $[0, 1)$, and a bounded function y satisfies

$$ry' + [c + rq(r)]y = e(r), \qquad e(r) = O(r),$$

then, for $0 < r < 1$,

$$y(r) = \frac{\displaystyle\int_0^r t^{c-1} \exp\left(\int_0^t q(s)\, ds\right) e(t)\, dt}{\displaystyle r^c \exp\left(\int_0^r q(s)\, ds\right)}. \tag{4}$$

Indeed, multiply by the integrating factor and integrate from a positive lower endpoint, then let it tend to zero. Boundedness and $c > 0$ annihilate the lower boundary term. In particular, a strictly positive (respectively negative) forcing throughout $0 < r < 1$ makes y strictly positive (respectively negative).

Lemma 1 (Common quadratic lower barrier)**.** *Let $a, b_1, b_2 > 0$, allowing $b_1 = b_2$, and suppose that $b_1 b_2 \le a(a + 1)$. Put*

$$c_i = a + b_i, \qquad A_i = \frac{ab_i}{c_i}, \qquad k = \frac{a^2}{c_1 c_2}.$$

Let $v(0) = 1$ and, for $0 < r < 1$, let $v(r) \in (0, 1)$ be the positive solution of

$$1 - v = \frac{kr}{1-r} v^2. \tag{5}$$

Then $f_{b_i}(r) > v(r)$ for $i = 1, 2$ and every $0 < r < 1$.

Proof. The explicit expression

$$v(r) = \frac{2}{1 + \sqrt{1 + 4kr/(1-r)}}$$

shows regularity at zero and positivity. Differentiating (5) gives

$$rv' = -\frac{(1-v)v}{(1-r)(2-v)}.$$

Introduce

$$\lambda = \frac{b_1 b_2 - a^2}{a}, \qquad T(r) = \frac{v}{(1-r)(2-v)}.$$

For either i, with $j \ne i$, direct simplification gives

$$c_i - \frac{A_i}{k} = a + b_i - \frac{b_i(a + b_j)}{a} = -\lambda.$$

Hence the residual of v in (3) is

$$E(r) := c_i(1-v) - \frac{A_i r}{1-r}v^2 - rv' = (1-v)\big(T(r)-\lambda\big). \tag{6}$$

We claim that $E(r) > 0$. If $b_1 b_2 \le a^2$, then $\lambda \le 0$ and the claim follows from $T > 0$. Otherwise $a^2 < b_1 b_2 \le a(a+1)$, so $0 < \lambda \le 1$ and

$$c_1 c_2 = (a+b_1)(a+b_2) \ge (a+\sqrt{b_1 b_2})^2 > 4a^2.$$

Thus $0 < k < 1/4$. Eliminating r using (5) now gives

$$T(r) - 1 = \frac{(1-v)(1-2kv)}{kv(2-v)} > 0. \tag{7}$$

It follows that $T(r) > 1 \ge \lambda$, even at the threshold $b_1 b_2 = a(a+1)$, proving the claim.

Set $D_i = f_{b_i} - v$. Subtracting the equation for v with residual (6) from (3) gives

$$rD_i' + \left[c_i + \frac{A_i r}{1-r}(f_{b_i} + v)\right] D_i = E(r).$$

Here D_i is bounded at zero and $E(r) = O(r)$. Formula (4), with $E > 0$, proves $D_i > 0$. □

Proof of Theorem 1. Strict parameter derivative. Suppose $b^2 \le a(a+1)$, and use Lemma 1 with $b_1 = b_2 = b$, so $k = a^2/c^2$. Differentiating (3) with respect to b, and using $\partial_b c = 1$ and $\partial_b A = a^2/c^2 = k$, yields, for $h = \partial_b f_b$,

$$rh' + \left[c + \frac{2Ar}{1-r}f_b\right] h = (1-f_b) - \frac{kr}{1-r}f_b^2.$$

The expression $y \mapsto 1 - y - kry^2/(1-r)$ is strictly decreasing for $y > 0$, and vanishes at v. Since $f_b > v$, the forcing is strictly negative. The function h is bounded at zero, and the forcing is $O(r)$. Equation (4) proves $h < 0$ throughout $(0,1)$.

Strict comparison of two parameters. Now suppose $b_1 < b_2$ and $b_1 b_2 \le a(a+1)$. Write $\Delta = b_2 - b_1 > 0$, $f_i = f_{b_i}$, and $W = f_1 - f_2$. We have $c_2 - c_1 = \Delta$ and $A_2 - A_1 = k\Delta$. Subtracting the two Riccati equations gives

$$rW' + \left[c_1 + \frac{A_1 r}{1-r}(f_1 + f_2)\right] W = \Delta\left[\frac{kr}{1-r}f_2^2 - (1-f_2)\right].$$

Lemma 1 gives $f_2 > v$, so the forcing is strictly positive. Formula (4) proves $W > 0$.

Necessity. The first terms of the defining series give

$$f_b(r) = 1 - \frac{ab}{(a+b)(a+b+1)}r + O(r^2).$$

Parameter differentiation therefore gives

$$\partial_b f_b(r) = \frac{a[b^2 - a(a+1)]}{(a+b)^2(a+b+1)^2}r + O(r^2).$$

If $b^2 > a(a+1)$, this is positive for sufficiently small positive r, proving necessity in (1). Similarly,

$$f_{b_1}(r) - f_{b_2}(r) = \frac{a(b_2 - b_1)[a(a+1) - b_1 b_2]}{c_1(c_1+1)c_2(c_2+1)}r + O(r^2).$$

If $b_1 b_2 > a(a+1)$, this is negative for sufficiently small positive r, proving necessity in (2). These local obstructions are also recorded in [1].

Finally, for $0 < a < 1$, both $b_1 = 1 - a$ and $b_2 = 3/2 - a$ are positive with $b_1 < b_2$, and

$$a(a+1) - (1-a)(3/2-a) = \frac{7a-3}{2}.$$

Equation (2) gives precisely the interval $[3/7, 1)$. □

Scope and status. This proves both assertions in the original source's corrected Conjecture 6.2. It also proves the more general pairwise criterion (2), with no separate campaign credit. The uncorrected conjecture, MathDB 364529, was excluded because the source already disproves it. Current MathDB and public-claim checks for 364530 are archived with the proof. The bounded literature audit located no matching full claim.

Task 2: Classify crossings outside the sharp region. For $b_1 b_2 > a(a+1)$, determine the number of zeros of $f_{b_1}(r) - f_{b_2}(r)$ in $0 < r < 1$, and characterize their dependence on the parameters.

Task 3: Determine the minimum with respect to the parameter. For fixed $a > 0$ and $0 < r < 1$, determine whether $b \mapsto f_b(r)$ has a unique minimum, and obtain sharp bounds and endpoint asymptotics for its location.

Task 4: Determine optimal uniform comparison bounds. For fixed $a, b_1, b_2 > 0$ with $b_1 < b_2$, determine $\inf_{0<r<1} f_{b_1}(r)/f_{b_2}(r)$, especially outside the region $b_1 b_2 \le a(a+1)$, where the lower bound 1 fails.

Task 5: Extend the sharp range to a shifted balance parameter. For fixed $\delta > 0$, classify the parameter-derivative and pairwise comparison ranges for

$$\frac{{}_2F_1(a, b; a+b+\delta; r)}{{}_2F_1(a, b; a+b; r)}, \qquad 0 < r < 1,$$

and determine which values of δ admit a scalar comparison equation analogous to (3).

References

[1] S.-L. Qiu, X.-Y. Ma and X.-Y. Xiang, *Turán-Type Inequalities for Gaussian Hypergeometric Functions, and Baricz's Conjecture*, arXiv:2408.15723v1, 28 August 2024.